



KSI
KUMARAGURU
SCHOOL OF
INNOVATION

24ADC001- BUSINESS INTELLIGENCE

2026-2027 LAB RECORD

**DEPARTMENT OF ARTIFICIAL
INTELLIGENCE AND DATA SCIENCE**

INDEX

Expt. No	Title of the Experiment	Page No	Total Marks	Faculty Signature
1.	Explore the interface and basic features of the BI tool			
2.	Load and visualize a sample dataset.			
3.	Import a dataset into the BI tool. And Cleanse data by handling missing values, outliers, and inconsistencies.			
4.	Transform data to suit BI reporting requirements and Design a dashboard with key performance indicators (KPIs).			
5.	Develop interactive dashboards for dynamic data exploration.			
6.	Integrate data from various sources for comprehensive analysis			
7.	Implement advanced chart types (treemaps, heatmaps, etc.).			
8.	Apply BI tools for forecasting and predictive analytics.			

Average marks

Faculty Signature

EXPT NO:1	EXPLORE THE INTERFACE AND BASIC FEATURES OF BI Tool
DATE: 10.07.2026	

PRE-LAB QUESTIONS

1. What is the purpose of Business Intelligence (BI) tools like Tableau and Power BI?

The primary purpose of BI tools is to transform raw, complex data into actionable, easy-to-understand visual insights. They help organizations:

- ♦ Connect to various data sources.
- ♦ Clean, model, and analyze data without needing deep programming knowledge.
- ♦ Create interactive dashboards that allow decision-makers to identify trends, spot anomalies, and make data-driven business decisions quickly.

2. List any three common data sources that Tableau and Power BI can connect to.

Tableau and Power BI are incredibly versatile and can connect to a wide array of sources, including:

- ♦ Flat Files: Microsoft Excel spreadsheets (.xlsx) and Comma-Separated Values files (.csv).
- ♦ Relational Databases: SQL Server, MySQL, or PostgreSQL.
- ♦ Cloud Warehouses / Services: Snowflake, Google BigQuery, or Salesforce.

3. What is a dashboard in BI, and how is it different from a single chart?

- ♦ A Single Chart (e.g., a line graph or a bar chart) visualizes a single data point, trend, or relationship. It answers a specific, narrow question (e.g., "How did stock prices trend last month?").
- ♦ A Dashboard is a curated collection of multiple charts, KPIs, and filters arranged on a single screen.
- ♦ The Difference: While a single chart gives you a single perspective, a dashboard provides a holistic, high-level overview of a business area. It is highly interactive, allowing users to filter all charts simultaneously to see how different metrics impact one another.

4. Define the following terms:

a) Visualization

The representation of data in a pictorial or graphical format (such as charts, maps, and graphs). It leverages human visual perception to make patterns, trends, and outliers instantly recognizable.

b) KPI (Key Performance Indicator)

A quantifiable measure used to evaluate the success of an organization, employee, or specific project in meeting performance objectives. In BI, a KPI is often displayed as a large, prominent number card.

c) Data Filter

A tool or rule used to restrict the data shown in a visualization. It hides irrelevant data points based on user-defined criteria (e.g. showing data only for the year 2025 or only for "Apple" stock).

5. Why is it important to clean and transform data before visualization?

Visualizing uncleaned data can lead to highly inaccurate charts and faulty business decisions. Cleaning and transforming data is critical because it:

- ♦ Fixes inconsistencies: Standardizes date formats (e.g., matching DD/MM/YYYY with MM/DD/YYYY) and text casing (e.g., merging "AAPL", "aapl", and "Apple").
- ♦ Handles missing/null values: Decides whether to remove blank fields or fill them with defaults (like 0 or the column average).

- ♦ Improves performance: Removing unnecessary rows or columns makes your dashboards load and respond much faster.

6. What kind of chart would you use to analyze stock price trends over time?

A Line Chart (or its specialized financial variation, the Candlestick Chart) is the best choice.

- ♦ Why: Line charts are designed to show continuous data points over time, making it incredibly easy to see upward or downward trends, volatility, and historical patterns at a glance.

7. Explain the role of slicers in Power BI and filters in Tableau.

Though named differently, Slicers (Power BI) and Filters (Tableau) serve the exact same purpose: they give the end-user control over what data is displayed.

- ♦ Instead of looking at a cluttered static dashboard, a user can click a slicer/filter (like a dropdown menu of Stock Names) to instantly slice the entire dashboard's data down to just the selected item.
- ♦ They turn a static report into an interactive playground for exploratory data analysis.

8. What does a calculated field or measure do in BI tools?

- ♦ Calculated fields (Tableau) and Measures (Power BI) allow you to create new data fields using formulas based on your existing data.
- ♦ Instead of manually altering your source Excel or SQL database, you can write mathematical, logical, or statistical formulas directly inside the BI tool to calculate metrics on the fly (e.g., calculating profit margin, date differences, or prediction errors).

9. What are the advantages of using interactive dashboards for financial data analysis?

- ♦ Real-Time Monitoring: Financial markets move fast; dashboards can be set up to refresh automatically with the latest market close data.
- ♦ Scenario Testing / What-If Analysis: Financial analysts can use parameters and slicers to test different scenarios (e.g., "What happens to our portfolio if this stock drops by 10%?").
- ♦ Drill-Down Capability: You can start at a high-level portfolio overview and click a specific stock to instantly "drill down" into its hourly performance and historical trends without opening a new file.

10. Which function would you use to find the absolute difference between predicted and actual stock prices?

You would use the ABS (Absolute Value) function.

- ♦ Formula: $\text{ABS}(\text{Predicted} - \text{Actual})$
- ♦ Why: It strips away any negative signs. If the prediction is off by \$5 (whether it's \$5 too high or \$5 too low), the error is still a positive \$5 difference, which is crucial for calculating average forecasting errors accurately.

IN-LAB EXERCISE:

Objective:

To understand the basic interface, data import process, and fundamental visualization capabilities of **Tableau** and **Power BI** using a stock prediction dataset, and to explore their usefulness in analyzing financial data.

Tools Required:

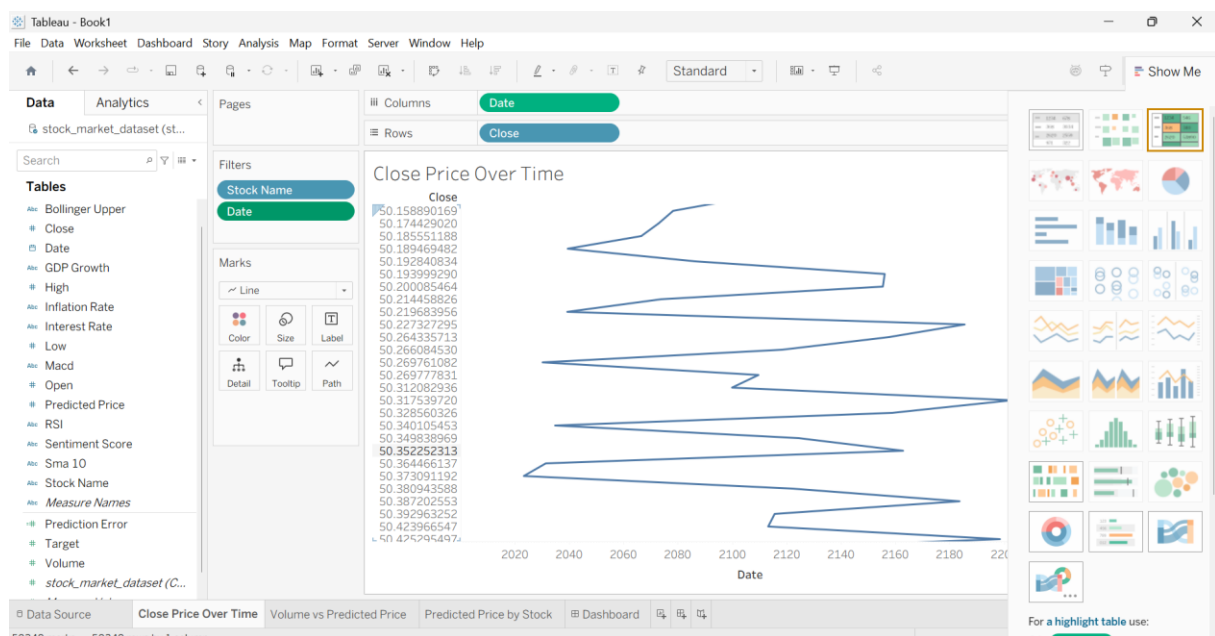
- Tableau Public / Tableau Desktop
- Power BI Desktop
- Sample Dataset (CSV/Excel) containing:
Date, Stock Name, Open, Close, High, Low, Volume, Predicted Price

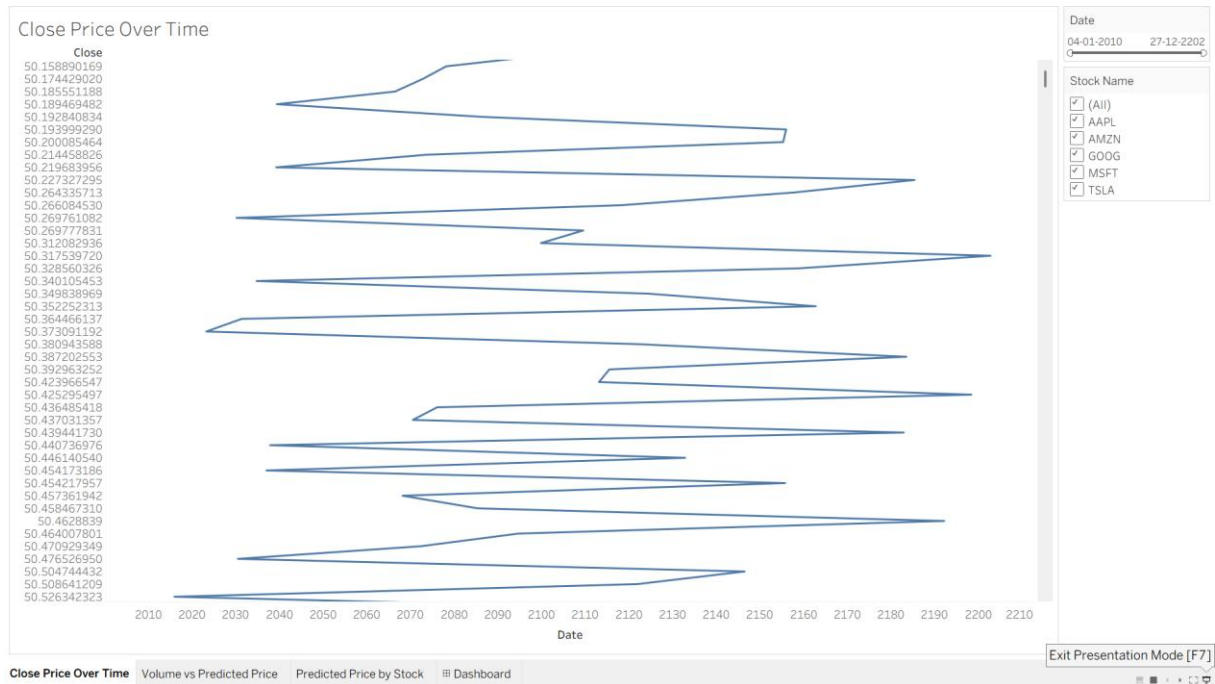
Exercise Steps:

Part 1: Working with Tableau

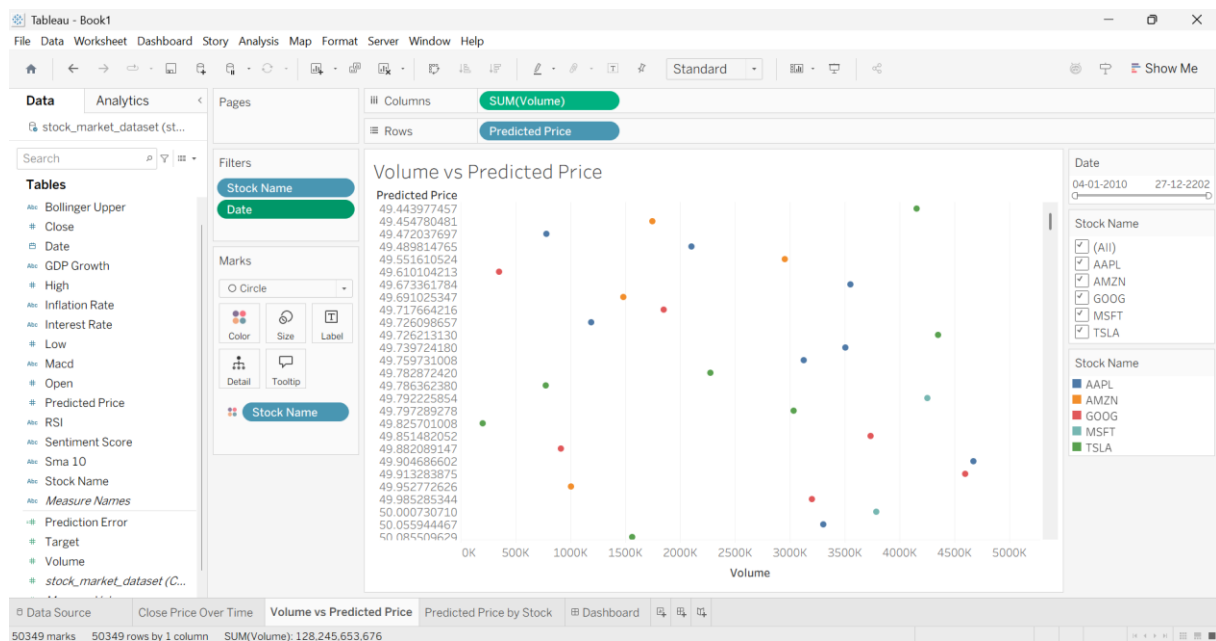
1. **Launch Tableau** and connect to the stock prediction dataset.
2. **Explore the Interface:**
 - Understand components like **Data Pane**, **Sheets**, **Marks Card**, and **Dashboards**.
3. **Create Basic Visualizations:**
 - **Line Chart:** Show Close Price over time.
 - **Scatter Plot:** Plot Volume vs. Predicted Price.
 - **Highlight Table:** Display Predicted Price by stock.
4. **Apply Filters:**
 - Add filters for Stock Name and Date Range.
5. **Create Calculated Field:**
 - Calculate prediction error as $|\text{Predicted} - \text{Close}|$.
6. **Design a Basic Dashboard:**
 - Combine your charts into one interactive dashboard.

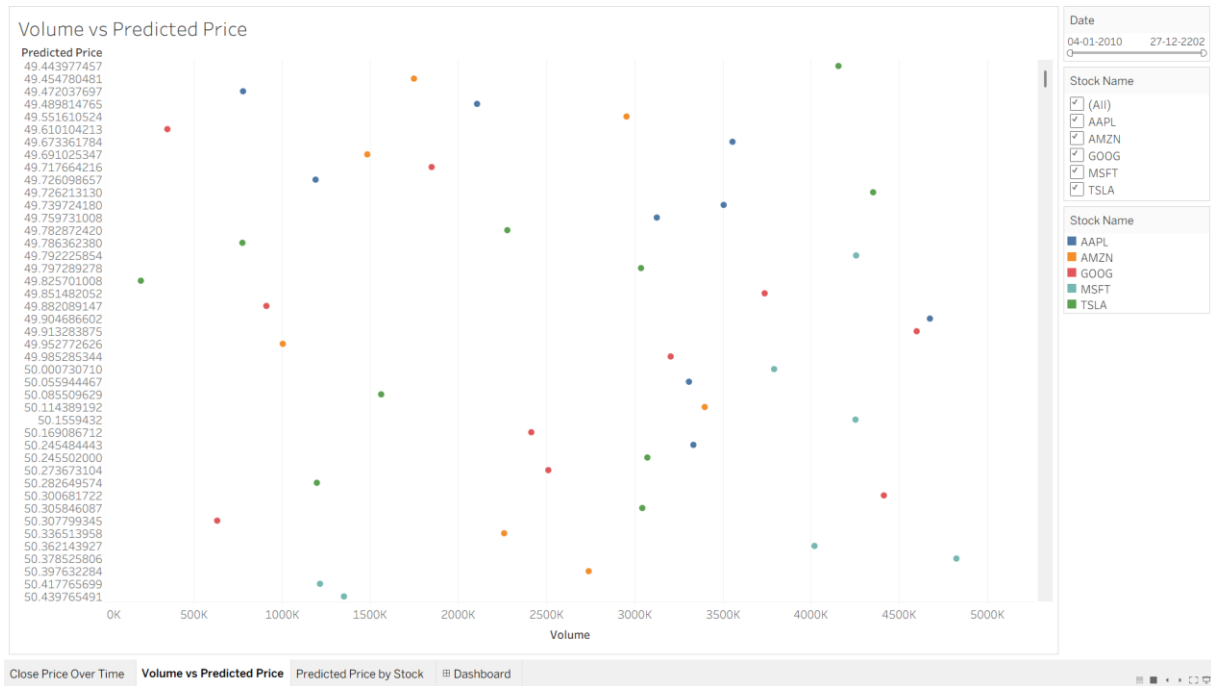
Line Chart(Close Price Over Time):



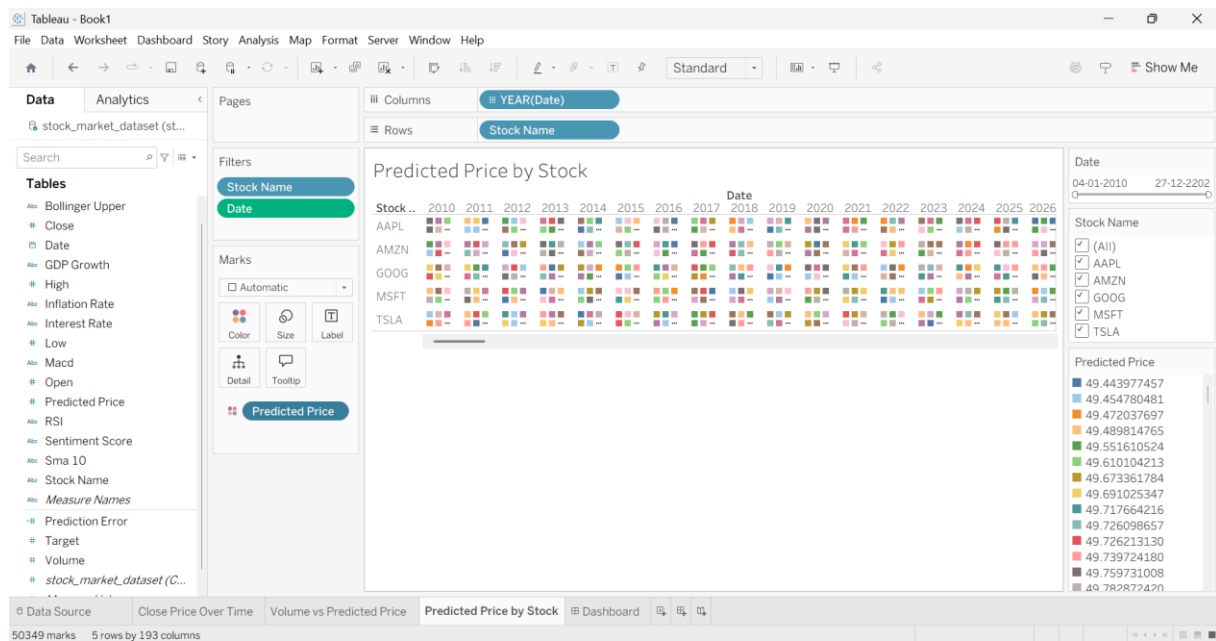


Scatter Plot(Volume vs. Predicted Price):



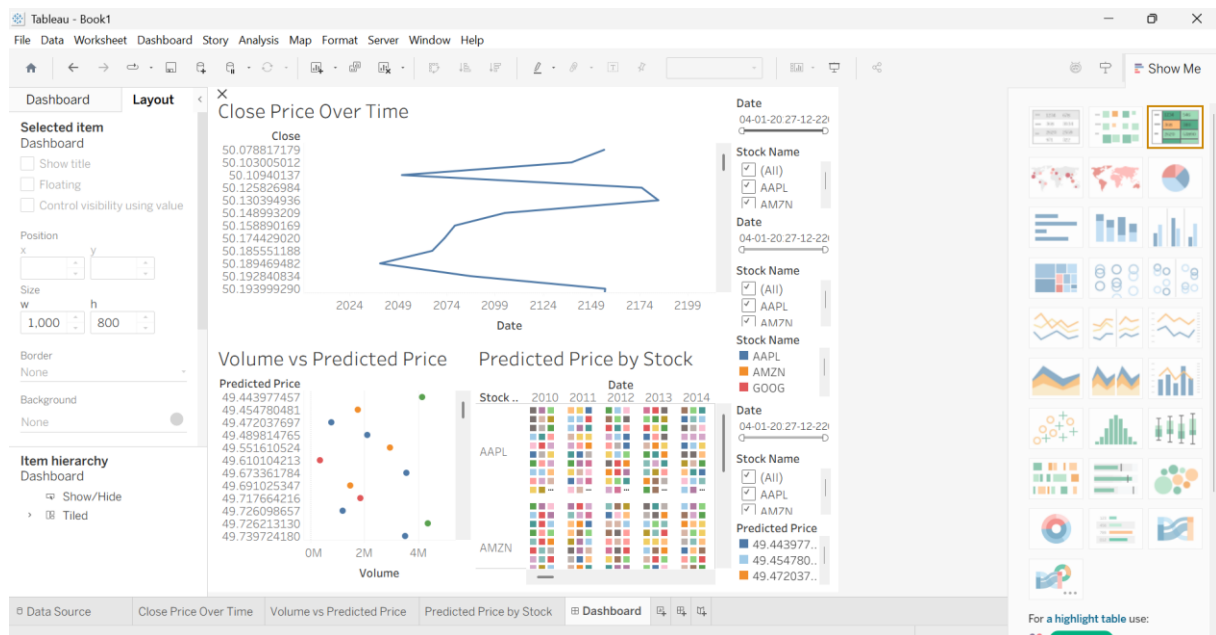


Highlight Table(Predicted Price by stock):





Dashboard:

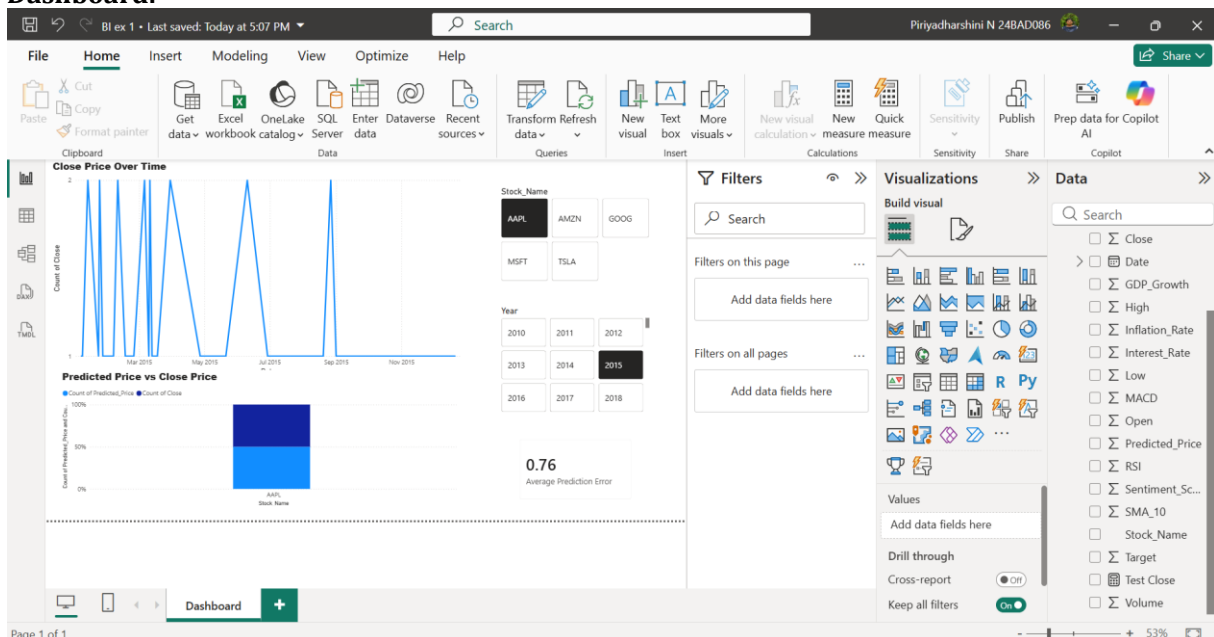




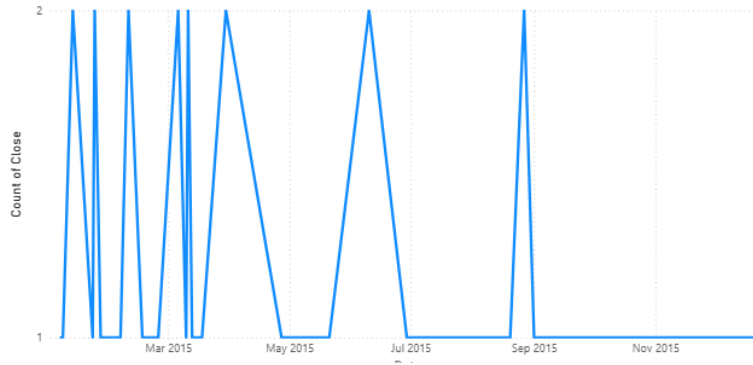
Part 2: Working with Power BI

1. **Open Power BI Desktop** and import the same dataset.
2. **Explore the Interface:**
 - o Familiarize with **Fields Pane**, **Visualizations Pane**, **Report View**, **Data View**.
3. **Create Basic Visualizations:**
 - o **Line Chart:** Trend of Close Price over time.
 - o **Bar Chart:** Compare Predicted Price vs. Close Price.
 - o **Card (KPI):** Display average prediction error.
4. **Use Filters and Slicers:**
 - o Filter data based on Stock Name and Date.
5. **Create a Simple Measure** using DAX:
 - o Example: $\text{Prediction Error} = \text{ABS}(\text{Predicted} - \text{Close})$
6. **Build a Report Page:**
 - o Arrange visuals in a single interactive page.

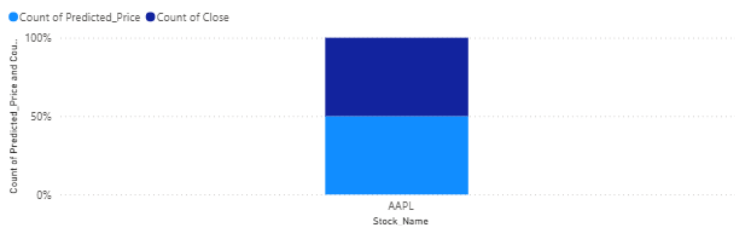
Dashboard:



Close Price Over Time



Predicted Price vs Close Price



Stock_Name

AAPL

AMZN

GOOG

MSFT

TSLA

Year

2010

2011

2012

2013

2014

2015

2016

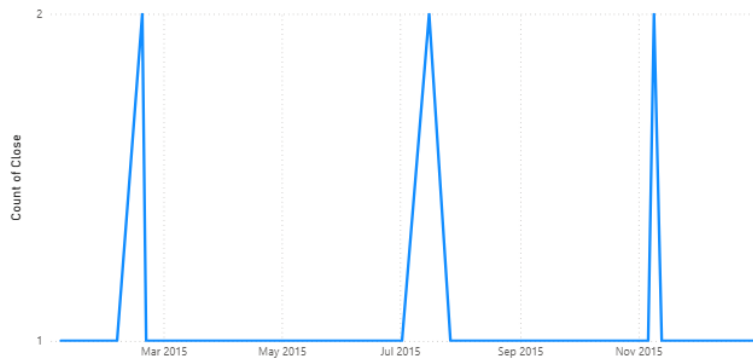
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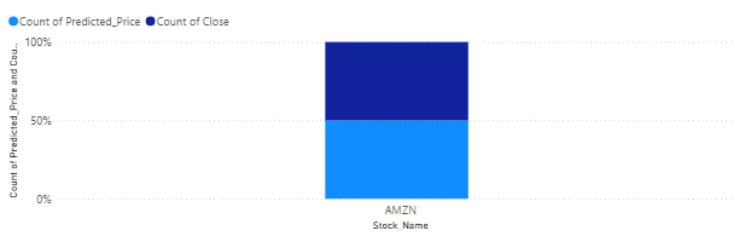
0.76

Average Prediction Error

Close Price Over Time



Predicted Price vs Close Price



Stock_Name

AAPL

AMZN

GOOG

MSFT

TSLA

Year

2010

2011

2012

2013

2014

2015

2016

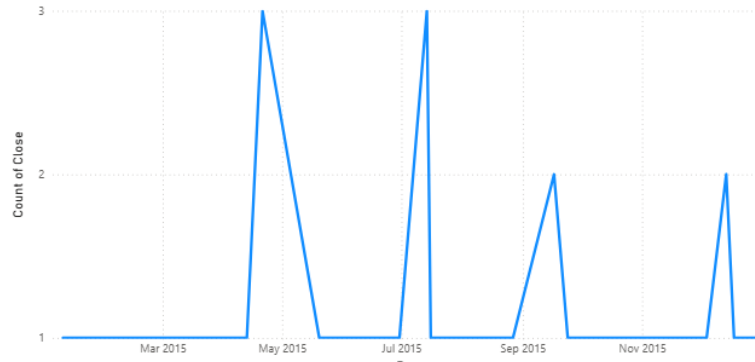
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2018

0.27

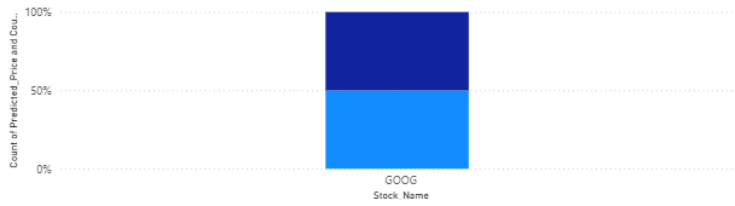
Average Prediction Error

Close Price Over Time



Predicted Price vs Close Price

● Count of Predicted_Price ● Count of Close



Stock_Name

AAPL

AMZN

GOOG

MSFT

TSLA

Year

2010

2011

2012

2013

2014

2015

2016

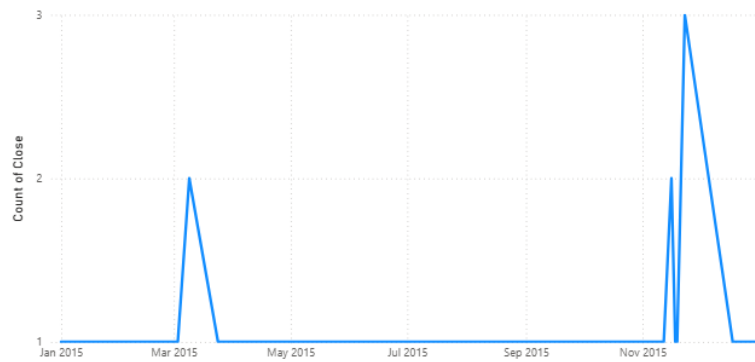
2017

2018

0.00

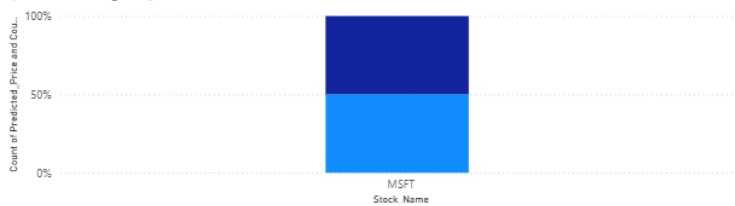
Average Prediction Error

Close Price Over Time



Predicted Price vs Close Price

● Count of Predicted_Price ● Count of Close



Stock_Name

AAPL

AMZN

GOOG

MSFT

TSLA

Year

2010

2011

2012

2013

2014

2015

2016

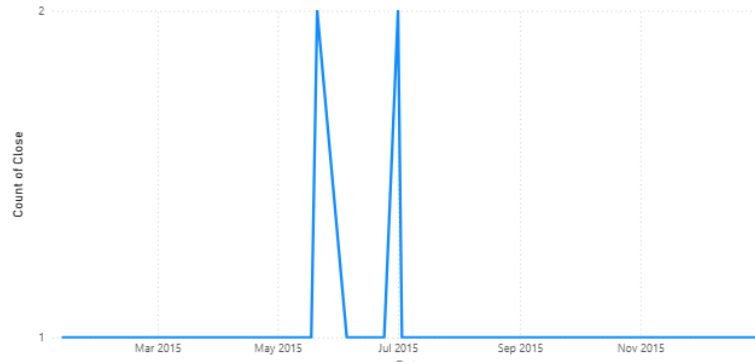
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2018

0.15

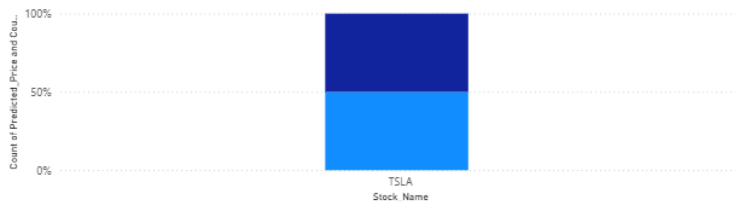
Average Prediction Error

Close Price Over Time



Predicted Price vs Close Price

● Count of Predicted_Price ● Count of Close



Stock_Name

AAPL

AMZN

GOOG

MSFT

TSLA

Year

2010

2011

2012

2013

2014

2015

2016

2017

2018

0.21

Average Prediction Error

POST-LAB QUESTIONS

1. Describe your experience with the Tableau interface. What components did you interact with?

The Tableau interface was easy to understand after some practice. I interacted with the Data Pane to access the dataset fields, the Columns and Rows shelves to create visualizations, the Marks Card to customize charts, the Filters shelf to filter data, the Show Me panel to select chart types, and the Dashboard workspace to combine multiple visualizations.

2. How did you import the dataset into Tableau and Power BI?

In Tableau, I clicked Connect → Text File, selected the CSV dataset, verified the data types, and opened a worksheet. In Power BI, I clicked Get Data → Text/CSV, selected the same dataset, previewed it, and clicked Load to import it into the report.

3. What types of visualizations did you create in both tools? List at least two per tool.

Tableau:

- ♦ Line Chart showing **Close Price over Time**
- ♦ Scatter Plot showing **Volume vs. Predicted Price**
- ♦ Highlight Table displaying **Predicted Price by Stock**

Power BI:

- ♦ Line Chart showing **Close Price over Time**
- ♦ Clustered Column Chart comparing **Predicted Price and Close Price**
- ♦ Card visual displaying the **Average Prediction Error**

4. How did you calculate the prediction error, and how was it displayed in your dashboard?

The prediction error was calculated as the absolute difference between the Predicted Price and the Close Price.

- ♦ Tableau Calculated Field:
ABS([Predicted Price] - [Close])
Power BI DAX Measure:
Average Prediction Error =
ABS(
AVERAGE(stock_market_dataset[Predicted_Price]) -
AVERAGE(stock_market_dataset[Close])
)

The prediction error was displayed using a **Card visual** in Power BI and could also be used in Tableau for analysis.

5. Which filtering or slicing techniques did you apply, and what insights did they reveal?

I applied Stock Name and Date filters (slicers in Power BI). These allowed me to analyze the performance of individual stocks over a selected time period. Filtering by a single stock made the line chart much clearer and helped compare predicted and actual closing prices more effectively.

6. Compare the process of creating a line chart in Tableau vs. Power BI. What differences did you notice?

In Tableau, creating a line chart involved dragging Date to the Columns shelf and Close Price to the Rows shelf, after which Tableau automatically generated the chart. In Power BI, I first selected the Line Chart visual and then assigned the Date and Close Price fields to the appropriate axes. Tableau was more drag-and-drop oriented, while Power BI required configuring the selected visual.

7. How did you organize the dashboard in Tableau and Power BI? What key elements did you include?

In Tableau, I combined the Line Chart, Scatter Plot, Highlight Table, and interactive filters into a single dashboard. In Power BI, I arranged the Line Chart, Clustered Column Chart, Prediction Error Card, and Stock Name and Date slicers on one report page to create an interactive financial dashboard.

8. Which tool (Tableau or Power BI) did you find more user-friendly for financial data visualization, and why?

I found Power BI more user-friendly because it provides a familiar interface, easy drag-and-drop visualizations, built-in DAX calculations, and interactive slicers. Tableau was excellent for creating attractive visualizations, but Power BI made it easier to build an interactive financial dashboard.

9. What challenges did you face while creating visualizations or dashboards?

I faced a few challenges, including incorrect data types after importing the dataset, choosing the correct aggregation for numerical fields, using the Card visual instead of the KPI visual in Power BI, and dealing with a cluttered line chart caused by a large dataset. These issues were resolved by correcting data types, adjusting aggregations, and applying filters.

10. How can the knowledge gained from this lab be applied to real-world financial data analysis?

The knowledge gained from this lab can be applied to real-world financial analysis by creating interactive dashboards to monitor stock performance, compare predicted and actual prices, identify market trends, calculate prediction errors, and support data-driven investment and business decisions using visualization tools like Tableau and Power BI.

ASSESSMENT

Description	Max Marks	Marks Awarded
Pre Lab Exercise	5	
In Lab Exercise	10	
Post Lab Exercise	5	
Viva	10	
Total	30	
Faculty Signature		